Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/31/2025
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Project Title:	PeleLM CFD of Ion Driven Winds from Diffusion Flames		
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Task Description:	The ultimate objective of this project is to comprehensively simulate the behavior of a small diffusion flame under the influence of an externally applied electric field in zero-gravity and to use PSI data to confirm the accuracy of that modeling. A detailed model will be used that includes the chemistry of charged ions and chemiluminescent flame intermediates to capture any feedback between ion-driven convection and combustion behavior, and to allow quantitative comparisons with experimental measurements. To date, the capability of accurately simulating electric field flames has eluded researchers because the system exhibits dramatic ranges of coupled temporal and spatial scales. Moreover, in earth gravity the hot combustion products are subject to buoyancy effects that are difficult to isolate from those generated by the electric field. We propose implementations to an existing powerful simulation framework (PeleLM) for this problem, and to use the PSI data set from E-FIELD Flames for validation of the model and investigation of the complex coupled system. PeleLM is a state-of-the-art reacting flow code tailored with a unique integration scheme that efficiently couples high-speed processes with the slower evolution of the flow structures using the adaptive mesh refinement (AMR) strategy. This project is to improve the ion chemistry (particularly employing reduced chemical mechanisms available in the published literature) to generalize the electric field configurations in PeleLM, and to then simulate zero-g coflow flames in the E-FIELD Flames dataset. Comparisons include V-I (voltage-ion current) curves and flame shapes (deduced from CH* images), varying with fuel types, compositions and flow rates. The data includes step function changes in voltage that can be used to evaluate the time response of the flame. It is important to also consider the impact of image and ion current collection timescales, which will be critical for model validation. The work takes advantage of the unique E-FIELD Flames data, measured in microgravity, to validate the proposed PeleLM model augmentations, to permit a detailed investigation of the ion dynamics under the influence of electric fields, and to evaluate the subsequent interactions with non-charged species, as well as to assess the potential of electric field forcing to affect thermal transport and sooting in diffusion flames. Once the simulation is validated against the zero-g data, it will be possible to further extend the simulations to flames in earth gravity, where potential applications for improved heat release and emission reduction can be explored. This utilization of zero gravity data from the E-FIELD Flames experiment for beneficial purposes on earth and for a better understanding combustion control is compatible with the NASA objectives. The E-FIELD Flames experiment was conceived and directed to create a data set that would permit the detailed assessment of the effects of ion-driven winds and charged species chemistry on non-premixed flames. A microgravity environment is required because the forces of ion driven winds and of gravity driven buoyancy winds are too close in magnitude to effectively separate their effects with any earth-gravity experiments where natural convection is a significant contributor to the combustion process. This rationale has been proven out by the E-FIELD flame experimental data now available as part of the PSI website. The Electric Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment was completed in 2018, and with nearly 150 hours of operations over six months and the ignition of 250 electric field flames on the ISS, it was a very successful measurement campaign. The experiment aspect of this project is to extract and assemble a data portfolio that is ideally suited for validating numerical simulations. The simulation aspect of the project employs the PeleLM framework at the National Renewable Energy Laboratory (NREL), which was previously developed in the Lawrence Berkeley National Laboratory (LBNL). This open-source code is for evolving chemically reacting flows at low Mach number with block-structured adaptive mesh refinement (AMR) using the AMReX library. It is the extension of this code that we exploit for comparison with E-FIELD Flames data, with the ultimate goal of understanding and then controlling hydrocarbon flames under the influence of electric fields. PeleLMeX is the non-subcycling version of PeleLM, and this project will use the non-subcycling strategy at the same time understanding other aspects using the PeleLM. Control of combustion using electric fields has for decades been proposed for extending flammability limits, reducing emissions, and preventing instability and blowoff, as well as modifying soot production. Implementation of the concept has been difficult, however, because of the complex coupling between buoyant driven flows, ion production in the flame, ion acceleration by the electric field, and the resulting forces on the neutral gas. With an efficient and accurate simulation capability, it will be possible to explore temporal and geometric variations of the electric field to manipulate flames in a wide range of beneficial ways. As the GRI-Mech 3.0-based super reduced mechanism was implemented but without excited species and with a few ions, in the prior fiscal year, the main focus of this cycle was to implement a more complicated San Diego-based reduced mechanism with positive ions, negative ions, and excited species. The specific aims and objective of this project in this year include: (1) Establishing flame images for fuel dilution cases, particularly for the 70% and 40% methane dilution in 1-g laboratory electric field flames (2) Understanding transient flame response, particularly the influence of the ion-driven wind in the 1-g in-laboratory electric field flames with the experimental flow conditions and mesh distance matching the microgravity E-FIELD Flames ISS experiments (3) Verify the strategy with understanding the detailed electric field with respect to the PeleLMeX function, and then implementing PeleLMeX CFD with a (newer) reduced chemical mechanism that includes more varieties of charged and excited species (4) Verify the fidelity of the PeleLMeX simulation of flames under the influence of electric fields by direct comparison using the previously validated reduced mechanism without electric fields in comparison to schlieren images of the flame under 1g conditions (5) Demonstrate validated PeleLM simulations of microgravity flames in steady conditions and during unsteady transitions in response to changes in the external electric field
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Once the simulation is validated against the zero-g data, it will be possible to further extend the simulations to flames in Earth gravity, where potential applications for improved heat release and emission reduction can be explored. This utilization of zero gravity data from the E-FIELD Flames experiment for beneficial purposes on Earth and for a better understanding combustion control is compatible with the NASA objectives.

As in the above objectives listing, this year has major aspects of establishing the 1-g laboratory microgravity flame dilution image data that resemble the International Space Station (ISS) conditions, optical diagnostic setup for understanding ion-driven wind flows using Schlieren imaging, testing the previous reduced mechanism in 1-g condition (which was tested in microgravity condition), and modifying the new reduced mechanism (with charged and excited species) with implementing the PeleLM CFD code for simulating flames under the influence of an electric field. This year, progress has been made on all the items (1-5) with 1, 2, 4, & 5 nearly completed. Specifically, the progress includes:
Computational Modeling Verification (with and without electric fields) — This task continues from the previous computational modeling effort in 0g ISS condition of E-FIELD Flames, using a reduced mechanism to predict ion concentrations in flames and the effects of ion driven winds in coflow. The focus is to simulate in gravity condition and observe how the computational fluid dynamics (CFD) simulation captures the difference between microgravity, using the E-FIELD Flames PSI dataset for effective validation of the computational models and 1g experiments. [Ed. Note: PSI is the NASA Physical Sciences Informatics (PSI) data repository for Physical Sciences experiments performed on the ISS.]

Experiment of Fuel Dilution and Optical Imaging Diagnostics in 1g using ISS Conditions — This test is to establish the methane fuel dilution part in laboratory experiments similar to the conditions on the ISS, understand the flame dynamics using Schlieren imaging, and further validate the results of the computational models described above.

Computational Modeling Implementation Employed with a Set of Species — To predict ion concentrations in flames and the effects of ion driven winds in coflow and jet flames, a complete set of charged species and excited species to include in the computation is essential and the most challenging. This task is to find the best scientific method to include more charged species and excited species implementation into the CFD simulation with appropriate reduced chemistry to ensure robust simulation capability.

The previous PeleLM simulation computation capability validation using pressurized conditions also continues. Further details of the project progress through the third year are described in the following.

PeleLMeX Computational Results and Discussion

The progress in the PeleLM computational aspects of the project are described in a series of conference papers provided in the publication section of this report and in recent American Society for Gravitational and Space Research (ASGSR) presentations (Grabon et al., 2024, Deu Morel et al., 2023). Much of the work has so far concentrated on the cases without an electric field in order to assess the complications of the code implementation, its conversion to cylindrically symmetric geometry, to identify the choice of reduced models, and to ensure that the boundary conditions are appropriate to simulate an extruded fuel tube.

Using PeleLM with Various Pressures and Fuel Dilution with Different Species – no electric field

This part of the progress continues to investigate high pressure methane diffusion flames with water addition in comparison to CO2 addition using PeleLM simulation (Esquivias et al., 2024). The study used the same geometry coflow burner as for E-FIELD Flames and examined its behavior under the pressure conditions of previous tests of water addition simulation. By examining a similar flame geometry, but under different conditions, the study continues to evaluate the potential for PeleLM simulation to capture the chemico-physical details of small diffusion flames. This work is to study the behavior of a methane diffusion flame with various amounts of water vapor and CO2 added to the fuel and compares the flame behavior with various percentages of water vapor and CO2 content (0% to 65%). It provides the detailed temperature, CO, and CO2 and other species concentration distribution using the PeleLM adaptive mesh refinement (AMR) code. This work computes the combustion behavior of a coflow burner flame jet using PeleLM. The flame was simulated to pressures of 1.0, 1.4, 5.7, and 11.1 atm. The results extracted and analyzed include temperature profiles and various species mole fractions compared with their equilibrium state. The fundamentals between water and CO2 addition on the flame behaviors, peak temperature, as well as maximum mole concentration of CO2 and CO, was studied. The details of how the peak values of species vary for water and CO2 addition as compared to their equilibrium values are investigated.

The peak concentration for species CO, CO2, OH, and H2O were plotted and compared between water addition and CO2 at two pressure conditions, 11.1 and 1 atm. In 1 atm, CO2 addition has less CO and H2O production in comparison to water addition, while the peak value of OH is generally higher in all of the CO2 addition cases. At 11.1 atm, OH is very similar with both diluents added, this is due to the compact nature of the flame under the high pressure that does not allow enough room for the thermo-chemical reaction to reveal itself. Because of the addition of CO2, there is a distinct reduction in the production of CO, meaning that water addition participated in the reaction as both chemical and physical effect, but again, these are the peak value comparison only at atmospheric and high pressure conditions. Detailed reaction evaluations still are needed, as well as a sensitivity analysis. In general, the effect of pressure can be seen in the increase of species of CO, OH, and H2O.

The species of O, H, and OH of water and CO2 addition were compared to equilibrium conditions. The ratio between the OH, O, and H mole fraction species and the mole fraction at equilibrium are plotted against with the increase of the addition concentration. The mole fraction at equilibrium is computed using the Colorado State University online calculator, which is based on the original Stanjan Equilibrium Code (Dandy, 2019). The comparison conditions are 0%, 30% and 65% water addition and CO2 addition to a stoichiometric fuel/air mixture. The ratio of, for example, Hthis sstudy/ Heq., presents the ratio between the equilibrium prediction and the kinetics result for these atoms at all pressure conditions. The results show that H and O are consistently more than two times their equilibrium expectation for all water content at lower pressure, while OH stays near the equilibrium value throughout all the conditions in this study.

There is also relatively limited effect of diluent composition, as the thermal images of flames diluted with water and carbon dioxide are very similar under similar pressure and dilution conditions. The addition of water and carbon dioxide has similar results on the flame behavior. Future work will include an analysis of existing high-pressure experiment results and the superequilibrium prediction for the species (H and O) at different pressures.

These details show that the PeleLM framework is able to capture the proper boundary conditions, even though the absolute geometry is slightly different, because the experiment has a slightly extruded fuel tube and the simulation has a flat inlet. The flow inlet is varied to approximate the conditions expected for the extruded case. Further results appear in all of the conference presentations and publications listed in the publications section of this report.

Preliminary Results 1g Electric Fields Using ISS Conditions – PeleLMeX

Task Progress:

The PeleLMeX computation of this research is implemented on the UCI High Performance Community Computing Cluster (HPC3) running on Linux 8.6. Its aim is to boost the effectiveness of simulations by utilizing a considerable number of cores. To handle user admission and resource allocation, the cluster employs entities like login nodes and the Slurm scheduler.

The goal of this project is to use the reduced mechanism Deu Morel et al. (2023, see reference) tested for microgravity condition with a newer version of the PeleLMeX as well as newer HPC3 hardware. This work initiated with reproducing the behavior of the coflow burner used in the ISS. The primary distinction between the numerical and experimental setups lies in the extruded fuel pipe. As the PeleLMeX code has not integrated the extruded geometry, it is unable to replicate the impact of the fuel pipe on the inflow velocities or temperature. The effect of the extruded tube is simulated using a customized inlet velocity profile. The primary assumption is that the Low Mach number regime is in effect. It is acknowledged that the velocities will remain within this regime. This is a widely accepted hypothesis in the study of combustion and was validated at the conclusion of the simulations.

The simulation was initiated with 19 (and 15) cm/s fuel flow velocity with 3.1 cm/s coflow air with 300K initial temperature with CFL 0.1. This case without the electric field can help explain the operation of PeleLMeX. Moreover, the cases without electric field are distinctive for shorter compilation time, due to the reduced number of equations/species to be solved resulting from the deactivation of the electric field. The reduced mechanism (by Di Renzo et al., 2022), comprising 26 species and 134 reactions with only two positive ions, H3O+ and HCO+, and one negative ion, O2- and electrons, was tested in microgravity and gravity conditions. This mechanism is currently used for simulating under 1g condition in comparison to the laboratory results to validate or invalidate the transient flame dynamic response to the ion wind in describing the E-FIELD Flames experiment.

As described in prior reports, for these simulations, the electric field implementation is in operation, requiring the use of HPC3, a supercomputer, due to its substantially longer computational time. Multiple convergence difficulties emerged during these simulations. After resolving these matters, we determined the necessary computation time to achieve a stable state. With the use of the electric field method, it takes 1 hour and 10 minutes to reach 1000 steps, in contrast to 3 minutes without it. Additionally, a stable state is achieved in the simulation within 60 milliseconds in simulation time. With the electric field, 1,000 steps correspond to a maximum of 10E-4 seconds. This requires over 30 days of consecutive simulations on the supercomputer to attain a steady state.

1g In-Laboratory Experiments

The 1g in-laboratory experiments include a coflow burner, rotameters for fuel and air, and a high voltage power supply (Trek 609A) with shunt current resistance for measuring ion current. The details of the experiment setups are described in Chien et al. (2018) Energies 11(5), 1235 and Chien et al. (2019) Combust. Flame 204(250-259). Both the High Voltage Power Supply (HVPS) output and ion current are monitored and controlled all along the running tests using data acquisition system NI 6341. The acquisition signals are sampled at a rate of 1000 samples per second and recorded with a LABVIEW code. The electrode potential is programmed before every recording allowing precise voltage sweeps. For safety, a High Voltage and Reverse Protection Circuit is used in addition to the HVPS. This specialized electronic arrangement is designed to protect electronic devices from potential damage caused by high voltage and reverse current. The High Voltage Protection circuit is composed of Zener diodes, which will open the circuit if the voltage reaches its threshold. This particular device safeguards the sensitive part of the circuit from being harmed by preventing electric arc formation, for instance. The Reverse Current Protection circuit is used to protect electronic components from being damaged by reverse polarity. If the circuit is incorrectly connected with the power supply by the user, diodes will block the current and avoid component destruction.

In this research, the mesh is set at 27 mm height above the burner due to the height limitation of the current experimental setup. The data collected includes ion current to applied voltages, natural flame images using a Nikon D90 camera, and also Shlieren imaging.

The measurements are made with two fuel flow speeds, 19 cm/s and 15 cm/s with airflow of 3.1 cm/s at 100% methane fuel concentration. The flame images were taken at ISO 200, aperture f/4.5, and a shutter speed of 1/10s, and the pictures were taken in the dark with a black fabric piece placed on the back to avoid reflections from the glass casing.

The ion current measurement is performed using a shunt resistor circuit. It consists of a 1 ¿Ω resistor connected in parallel with two Zener diodes. This circuit is symmetrical, which means the polarity isn’t a concern here. The ion current results for positive electric fields are measured and recorded as a confirmation. The results indicate that a positive E-FIELD leads to the appearance of three regions. In the sub-saturation region, ion current increases with applied voltage until it reaches a plateau, known as the saturation region, which is reached at around 1.65 kV/cm for each flame speed. Finally, there is a final increase in ion current in the supersaturation region. The limits between the saturation regions are not the same depending on the flame velocity due to the total amount of carbon content input. Therefore, the saturation plateau appears at a lower electric field strength for lower velocity flames.

To understand how the ion-driven wind and the process of electric fields affect the flame dynamics, the Shlieren imaging optics diagnostic technique is performed. The Schlieren imaging is an optical setup composed of two parabolic convergent mirrors (f=48’’), a punctual light source, a razor blade and the Nikon D90 camera as the focal plane. The Z-type optical setup is used due to space limitations. The principle of Schlieren imaging is that the light rays reflected by the first mirror can be considered parallel because the light source is placed at its focal distance. If nothing is placed between the mirror, the light continues its path through the system and the image projected on the camera sensor is blank. However, if a change in density, which implies a change of refraction index, occurs between the two mirrors, then the ray light will not arrive parallel in the second mirror. These rays will then either be blocked by the razor blade and appear darker or pass above the razor blade and appear brighter.

Schlieren imaging is efficient when it comes to observing density changes and will be employed to validate the simulation. The changes in the flame shape are more remarkable on the 19 ¿¿/¿ flame in comparison to the lower flow rate. The Schlieren imaging also shows that the thermal plume is wider for the higher flow rate. As the electric field strength increases for the positive field, the flame is concentrated and pulled upward to the mesh and the thermal plume is narrower. For the negative field configuration, the flame is pushed toward the burner and becomes wider; the thermal plume would be widened instead. When the saturation plateau is reached at around 1. 6 ¿¿/¿¿, the flame shape doesn’t change significantly, nor does the thermal plume as the electric field strength increases. The electric field can affect the soot production (orange region of the flame), which can be a great potential for application as soot production can

indicate a more efficient combustion.

[Ed. Note: Per the Principal Investigator (PI), the investigation gathered images of negative field and positive field with fuel dilution at 70% and 40%. These images resemble the fuel dilution conditions from the 0g experiments, and further analysis is underway.]

Comparison Between Simulation and Experiments

An initial comparison of the Schlieren imaging and the density profile from the PeleLMeX computation, specifically for flame widths for both flame velocities, are conducted. The result is computed from the reduced mechanism (Di Renzo 2022) under gravitational conditions, which was tested in microgravity. We understand that the comparison is qualitative as each image is normalized relative to its maximum intensity. A detailed Schlieren imaging is needed as well as a higher acquisition rate, for capturing the images between electric field charges can help further understand flame temperature and fully authenticate the chemical mechanism including more positive, negative ions and excited species from the PeleLMeX. The work is currently being carried out on a local high performance computing cluster, but there are limited hours available for that system so additional computational resources will be explored. Further results appear in the publications identified in that section of this report.

Preliminary Outcome for 0g Implementation with More Species – PeleLMeX

To bridge the gap between simulation outcomes and empirical data, this part of the work is to evaluate potential mechanism candidates: (A) 30 species and 86 reaction using Coffee mechanism as foundation (1993) by Pederson and C. Brown; (B) 26 species and 134 reactions but only 4 ionic species using GRI Mech 3.0 by Renzo et al.; and (C) the recently published 45 species and 216 reactions using San Diego reduced mechanism, by López-Cámara et al. This task is to implement the recently published reduced San Diego mechanism in microgravity. These mechanisms are chosen for their comprehensive coverage of ionic and exited species within the flame, compared to the mechanism currently used by Di Renzo et al. which is not able to capture the ion current behavior in microgravity condition as part of work from the previous fiscal year.

The main difference lies in the (C) reduced San Diego mechanism chemical model, which includes 11 charged species, including H3O+, HCO+, C2H3O+, CH5O+, O2−, OH−, e−, CO3−, CHO2−, O−, and CHO3− (in comparison to the previous used (B) H3O+, HCO+, O2− and e−) and included excited species CH and OH. This mechanism includes many species, and as the electric field is introduced, the calculation became extensive and the computation time increased substantially. With various tests and approaches for launching the mechanism at various stages, the simulation remains computing with extended calculation time, unresolved and/or unstable with inconsistent results. The heat release from the calculation can be massive at a single cell and that can lead to a relatively large local flow, and/or the chemical solver may fail to sort through the reaction and lead to confusion. Further tests on the mechanism, particularly on sorting through how the solver processes each step in the calculation, as well as how the mechanism works with a relatively newer version of the simulation, are needed.

Conclusions

The modeling work is progressing with the newer version of PeleLMeX using the reduced mechanism (with four ions, electron, and no excited species) in gravitational environment. The experiments with diluents are completed, and the ion current data and the Schlieren imaging are verified using ISS conditions. Locally with PeleLM non-subcycling version, the code has also continued to be exercised under different coflow flame conditions with different levels of fuel dilution with carbon dioxide and different pressures to demonstrate the validity from the previous work with water addition of the boundary condition implementation. Challenges remain for how to implement a relatively large San Diego based mechanism including more ions and charged species into the newer version of PeleLMeX. Continuing work on this project focuses more heavily on the computations with detailed assessment of the ion current prediction and the role the chemical mechanism plays in that prediction.

References

Esquivias, B. and Chien, Y.-C. (2024), “A comparison between water addition and CO2 addition to a diffusion jet flame,” Combustion Science and Technology, Special issue of the 29th ICDERS. doi: 10.1080/00102202.2024.2380083

Deu Morel, R., Day, M. and Chien, Y.-C. (2023) “Investigation of mechanisms for microgravity ion-driven wind using simulation,” Graduate Poster Presentation, American Society for Gravitational and Space Research Annual Meeting 2023, Nov. 14 –18, Washington DC.

Grabon, K., Wimer, N., Deak, N., Day, M, and Chien, Y.-C. (2024) “Ion-driven wind of E-FIELD Flames,” Combustion 2 Presentation, American Society for Gravitational and Space Research Annual Meeting 2024, Dec. 3 – 7, San Juan, Puerto Rico.

Invited Technical Lectures/Presentations

NASA Physical Sciences Informatics (PSI) User Group, “ACME — E-FIELD Flames,” April 11th, 2024 (Virtual).

Women in Technology at UCI (WITIN) inclusive networking, “NASA Keck, Diversity & Lovin’ UCI,” Irvine, California. May 17th, 2024 (Virtual).

Bibliography Type:	Description: (Last Updated: 11/10/2025)
Abstracts for Journals and Proceedings	Grabon K, Wimer N, Deak N, Day M, Chien, Y-C. "Ion-driven wind of E-FIELD flames." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Abstracts for Journals and Proceedings	Chien Y-C. "ACME E-FIELD flames: Microgravity electric field flames." 17th Southern California Flow Physics Symposium (So Cal Fluids XVII), Irvine, California, April 13, 2024. Abstracts. 17th Southern California Flow Physics Symposium (So Cal Fluids XVII), Irvine, California, April 13, 2024. , Apr-2024

Abstracts for Journals and Proceedings	Chien Y-C, Deu Morel R, Dru L. "Microgravity E-FIELD flames simulation using PeleLMeX." 40th International Symposium on Combustion, Milan, Italy, July 21-26, 2024. Abstracts.40th International Symposium on Combustion, Milan, Italy, July 21-26, 2024. Work in Progress Poster. , Jul-2024
Articles in Peer-reviewed Journals	Esquivias B, Chien,Y-C. "A comparison between water addition and CO2 addition to a diffusion jet flame." Combustion Science and Technology. 2024 Jul 24;196(13):2170-86. Special Issue of the International Colloquium on the Dynamics of Explosions and Reactive Systems (ICDERS). https://doi.org/10.1080/00102202.2024.2380083 , Jul-2024
Dissertations and Theses	Grabon K. "Electric field effect on non-premixed methane flames under 1g conditions and PeleLMeX simulation using microgravity conditions." M.S. Thesis for Engineering Internship from ISAE-ENSMA, University of California, Irvine, December 2024. , Dec-2024
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